Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

[18EEOE03]

SEVENTH SEMESTER B. Tech EXAMINATION, DECEMBER - 2022

OPERATIONS RESEARCH

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Define operations research. Explain the different phases of operations research **07**

b. Use the graphical method to solve the following LP problem. **07**

Maximize $Z = 2x1 + x2$

subject to the constraints

i. $x_1 + 2x_2 \leq 10$,

ii. $x_1 + x_2 \leq 6$,

iii. $x_1 - x_2 \leq 2$,

iv. $x_1 - 2x_2 \leq 1$ and

v. $x_1, x_2 \geq 0$

2. a. Find Solution using Vogel's Approximation method, also find optimal solution using MODI (u-v) method **10**

	D1	D2	D3	D4	Supply
S1	11	13	17	14	250
S2	16	18	14	10	300
S3	21	24	13	10	400
Demand	200	225	275	250	

b. Differentiate between transportation model and assignment model. **04**

3. a. Solve the game with the following pay-off matrix for firm A. Using dominance property to reduce the matrix to 3 X 2 and further solve the 3 X 2 matrix graphically. **09**

		Player B			
		1	2	3	4
Player A	1	3	2	4	0
	2	3	4	2	4
	3	4	2	4	0
	4	0	4	0	8

b. What is a game? Explain its characteristics. **05**

OR

4. a. Define pure strategy and mixed strategy in the game theory with examples **04**

b. Consider a modified form of "matching biased coins" game problem. The matching player is paid Rs. 8.00 if the two coins turn both heads and Rs. 1.00 if the coins turn both tails. The non-matching player is paid Rs. 3.00 when the two coins do not **05**

match. Given the choice of being the matching or non-matching player, which one would you choose and what would be your strategy?

c. Find the optimal plan for both the player and also calculate the value of game. **05**

		PLAYER B			
		I	II	III	IV
PLAYER A	I	-2	0	0	5
	II	4	2	1	3
	II	-4	-3	0	-2
	IV	5	3	-4	2

5. a. Explain the following terms **05**

a. Earliest time

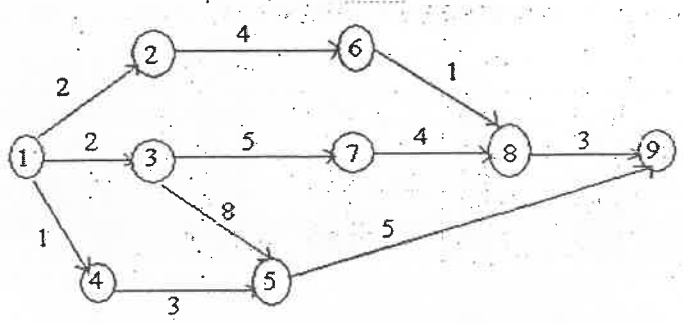
b. Latest time

c. Total activity slack

d. Event slack

e. Critical path

b. Find the critical path and calculate the different time estimates for the following network **09**



OR

6. a. Explain the basic steps in PERT/CPM techniques. **04**

b. A project has the following characteristics **10**

Activity	Most optimistic time	Most pessimistic time	Most likely time
(1 – 2)	1	5	1.5
(2 – 3)	1	3	2
(2 – 4)	1	5	3
(3 – 5)	3	5	4
(4 – 5)	2	4	3
(4 – 6)	3	7	5
(5 – 7)	4	6	5
(6 – 7)	6	8	7
(7 – 8)	2	6	4
(7 – 9)	5	8	6
(8 – 10)	1	3	2
(9 – 10)	3	7	5

Construct a PERT network. Find the critical path and variance for each event. What is the probability that the jobs on the critical path can be completed in 31 days?

7. a. Explain the different types of forecasting in decision making **08**

b. Alpha company has the following sales pattern. Compute to sales forecast for the year 10. **06**

Year	1	2	3	4	5	6	7	8	9
Sales (in Lakhs)	6	8	11	23	29	34	40	45	56

OR

8. a. What are the types of demand pattern? Explain them with suitable sketches **08**

b. Given below is a series of weekly demand data the Beta Company collected on one of its products and forecasts for the corresponding weeks, made by a forecast method which the company is testing. **06**

i. Compute the mean absolute deviation based on all six weeks of data

ii. Compute the mean squared error based on six weeks data

Week	1	2	3	4	5	6
Demand	142	181	144	174	192	176
Forecast	155	157	159	161	163	165